# MSG No. 121.331 $I\bar{4}2'm'$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `-42'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 2: Wyckoff site: `2b`, site symmetry: `-42'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 3: Wyckoff site: `4c`, site symmetry: `22'2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[1,2,5,6]` |
| 2 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | `[3,4,7,8]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[9,10,13,14]` |
| 4 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[11,12,15,16]` |

Table 4: Wyckoff site: `4d`, site symmetry: `-4..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,2,11,12]` |
| 2 | $\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$ | `[3,4,9,10]` |
| 3 | $\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[5,6,15,16]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$ | `[7,8,13,14]` |

Table 5: Wyckoff site: `4e`, site symmetry: `2.m'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,2,7,8]` |
| 2 | $[0,\ 0,\ -z]$ | `[3,4,5,6]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,10,15,16]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[11,12,13,14]` |

Table 6: Wyckoff site: `8f`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,5]` |
| 2 | $[-x,\ 0,\ 0]$ | `[2,6]` |
| 3 | $[0,\ -x,\ 0]$ | `[3,7]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 4 | $[0,\ x,\ 0]$ | `[4,8]` |
| 5 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[9,13]` |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[10,14]` |
| 7 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | `[11,15]` |
| 8 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | `[12,16]` |

Table 7: Wyckoff site: `8g`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | `[1,5]` |
| 2 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | `[2,6]` |
| 3 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | `[3,7]` |
| 4 | $\left[0,\ x,\ \frac{1}{2}\right]$ | `[4,8]` |
| 5 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[9,13]` |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | `[10,14]` |
| 7 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | `[11,15]` |
| 8 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$ | `[12,16]` |

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,2]` |
| 2 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | `[3,4]` |
| 3 | $\left[0,\ \frac{1}{2},\ -z\right]$ | `[5,6]` |
| 4 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[7,8]` |
| 5 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$ | `[9,10]` |
| 6 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[11,12]` |
| 7 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$ | `[13,14]` |
| 8 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[15,16]` |

Table 9: Wyckoff site: `8i`, site symmetry: `..m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | `[1,8]` |
| 2 | $[-x,\ -x,\ z]$ | `[2,7]` |
| 3 | $[x,\ -x,\ -z]$ | `[3,5]` |
| 4 | $[-x,\ x,\ -z]$ | `[4,6]` |
| 5 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,16]` |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$ | `[10,15]` |
| 7 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$ | `[11,13]` |
| 8 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[12,14]` |

Table 10: Wyckoff site: `16j`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | `[1]` |
| 2 | $[-x,\, -y,\, z]$ | `[2]` |
| 3 | $[y,\, -x,\, -z]$ | `[3]` |
| 4 | $[-y,\, x,\, -z]$ | `[4]` |
| 5 | $[x,\, -y,\, -z]$ | `[5]` |
| 6 | $[-x,\, y,\, -z]$ | `[6]` |
| 7 | $[-y,\, -x,\, z]$ | `[7]` |
| 8 | $[y,\, x,\, z]$ | `[8]` |
| 9 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | `[9]` |
| 10 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | `[10]` |
| 11 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | `[11]` |
| 12 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | `[12]` |
| 13 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | `[13]` |
| 14 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | `[14]` |
| 15 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | `[15]` |
| 16 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | `[16]` |